ABID AHMED SHAIKH

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PROFESSIONAL SUMMARY

Information Technology undergraduate with hands-on experience in software engineering, backend systems, cloud-native infrastructure, and AI-integrated application development. Currently working as a Software Engineer Intern at Bitkraft Technologies LLP, contributing to AR-based game development, scalable backend architecture, runtime content delivery systems, and developer tooling using Unity, AWS, Next.js, Node.js, and TypeScript. Experienced in experimenting with AI pipeline architectures for diffusion model training, persistent-state workflows, and scalable video generation approaches across constrained compute environments. Co-author of two research papers accepted at ICSSSD 2026, including a Best Paper Award-winning publication on agentic effort intelligence systems for real-time productivity analysis. Skilled in cloud infrastructure, microservices, CI/CD pipelines, Generative AI, and full-stack application development across AWS and Oracle Cloud ecosystems.

EXPERIENCE

Software Engineer Intern

Bitkraft Technologies LLP [23rd Feb, 2026 - Present]

AR & Game Engineering

- Fixed critical AR issues including **GPS instability** and **portal drifting** in a Unity-based AR game, significantly improving **gameplay stability** and **spatial accuracy**.
- Integrated and optimized **10 Unity mini-games** from commercial asset packs, improving **gameplay responsiveness** and **interactive performance**.
- Developed a custom **360° space shooter mini-game** featuring **dynamic enemy spawning** and enhanced gameplay mechanics.
- Implemented **dynamic scene loading** using **Unity Addressables** and **AWS S3**, enabling **modular runtime content delivery** and scalable game updates.

Backend & Platform Engineering

- Re-architected the **legacy PHP backend** into a modern backend system using **Next.js, Node.js, and TypeScript** to improve **scalability** and **maintainability**.
- Designed and developed a **no-code/low-code game authoring system** enabling non-developers to configure game logic and workflows without manual JSON editing.
- Built an **asset management system** for direct media uploads to **AWS S3** through the backend dashboard, eliminating manual cloud upload dependencies.
- Reverse-engineered and documented the existing **MVC architecture** while preserving **legacy system behavior** and compatibility.
- Established **GitHub-based SDLC workflows** including GitHub Projects, issue tracking, structured commits, and collaboration workflows.
- Authored **technical documentation**, onboarding guides, and system manuals for Unity and backend systems.

AI Pipeline Engineering

- Proposed and experimented with an **infrastructure-independent diffusion model training pipeline** designed to preserve **intermediate training states** in **persistent storage** rather than volatile memory.
- Explored **low-resolution video generation workflows** using image-based generation pipelines combined with **temporal super-resolution (TSR)** techniques for scalable video synthesis across constrained compute environments.

- Implemented components of automated **web** and **YouTube scraping pipelines** for collecting and preprocessing **video datasets** used in diffusion model experimentation workflows.

RESEARCH PUBLICATIONS

Cognitrace: An Effort Intelligence System for Real-Time Cross-Context Productivity Analysis
Co-author | Accepted at ICSSSD 2026 (Paper ID: 319)

- Proposed an **agentic effort intelligence framework** for analyzing productivity patterns using **real-time behavioral analytics** and **cross-context activity analysis**.
- Contributed to research focused on intelligent productivity monitoring, contextual behavioral interpretation, and adaptive effort evaluation systems.
- Awarded the **Best Paper Award** at ICSSSD 2026.

Green Passport: An Agentic AI-Verified Digital Product Passport for Greenwashing Mitigation
Co-author | Accepted at ICSSSD 2026 (Paper ID: 196)

- Proposed a **digital product passport framework** integrating **AI-assisted sustainability claim verification**, **semantic analysis**, and **cryptographic audit mechanisms**.
- Contributed to research focused on transparency, sustainability verification, and AI-driven validation of environmental claims.

PROJECTS

Ascend APEX - Academic Platform for Excellence (Hackathon Winner)

- Built a full-stack academic platform using **Oracle APEX** and **OCI Generative AI**.
- Integrated **RAG-based retrieval**, Serp API jobs, and **Oracle APEX workflows**.
- Won the **National Oracle APEX Hackathon Xcelerate25** (₹100,000).

Microservices and CI/CD Pipeline Builder

- Migrated a monolithic app to **Docker-based microservices**.
- Built an automated **CI/CD pipeline** using AWS CodeCommit, CodePipeline, and CodeDeploy.
- Implemented **blue-green deployments** with container images stored in **Amazon ECR**.
- Deployed services on ECS Fargate with an **Application Load Balancer** and 4 target groups.

Cloud Web Application Builder

- Built a fault-tolerant web system with **AWS EC2**, **RDS**, and **VPC**.
- Implemented **Auto Scaling** and an Application Load Balancer to ensure high availability.
- Managed secrets using **AWS Secrets Manager** and produced comprehensive documentation.

TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, TypeScript
- **Backend & Web Development:** Node.js, Next.js, Oracle APEX, React.js, Express.js
- **Cloud & Distributed Systems:** AWS, Oracle Cloud Infrastructure (OCI), Containerized Systems (Docker, ECS), Microservices Architecture, Serverless & Scalable Applications
- **DevOps & Infrastructure Engineering:** CI/CD Pipelines, Jenkins, Terraform, Git, GitHub, Infrastructure Automation, Cloud Deployment Pipelines
- **AI/Machine Learning/Generative AI:** AI Agent Systems, RAG (Retrieval-Augmented Generation), Prompt Engineering, Diffusion Models (Experimental), Semantic Analysis, AWS Bedrock, Amazon SageMaker
- **Game Development & Real-Time Systems:** Unity Engine, Augmented Reality (AR), Runtime Content Delivery Systems, Unity Addressables
- **Tools & Developer Productivity:** GitHub Copilot, OpenAI Codex, AntiGravity, Kiro

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** – Amazon Web Services
- **Oracle Cloud Certifications** (6x): OCI Foundations, AI Foundations, Data Platform Foundations, Generative AI Professional, AI Vector Search Professional, APEX Developer Professional
- **Red Hat Academy:** RH124 System Administration I
- **CodeSignal**: Prompt Engineering for Everyone, Mastering Cloud Engineering with AWS & Python, Building Generative AI Applications with AWS, Enterprise Machine Learning with Amazon SageMaker

LEADERSHIP & ACHIEVEMENTS

- **President, AWS Academy – MHSSCOE (2025-26)**
- **Joint Head, ACM Editorial Team (2025-26)**
- **Winner, National Oracle APEX Hackathon Xcelerate25** – ₹100,000 prize
- **AWS STJ Prompt Engineering Competition** – Score: 533/600
- **Completed advanced training in cloud engineering, AI/ML, and prompt engineering**, applying these skills to projects in AWS, Oracle Cloud, and AI-integrated applications

EDUCATION

B.E. Information Technology - M.H. Saboo Siddik College of Engineering (Mumbai University)
Expected 2027 | CGPA: 8.96